IoT Challenge 2

Packet Sniffing Exercises

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Exercise Question 1 (EQ1)

The temperature sensor transmits one measurement every 2 minutes, so over 1 hour the total number of measurements is:

$$N = \frac{60}{2} = 30$$

The radio energy costs are:

$$E_{TX} = 50\,\text{nJ/bit}, \qquad E_{RX} = 58\,\text{nJ/bit}$$

Since 1 byte = 8 bits, the energy for transmitting or receiving a message of B bytes is:

$$E_{tx}(B) = B \cdot 8 \cdot 50\,\text{nJ}$$

$$E_{rx}(B) = B \cdot 8 \cdot 58\,\text{nJ}$$

(a) CoAP direct communication

For each measurement, CoAP uses:

- one **CON PUT Request** of 70 B from sensor to actuator
- one **PUT ACK** of 15 B from actuator to sensor

Thus, for one CoAP exchange:

Sensor energy per exchange The sensor transmits the PUT request and receives the ACK:

$$\begin{aligned} E^{\text{CoAP}}_{\text{sensor},1} &= E_{tx}(70) + E_{rx}(15) \\ &= 70 \cdot 8 \cdot 50 + 15 \cdot 8 \cdot 58 \\ &= 28000 + 6960 = 34960\,\text{nJ} = 34.96\,\mu\text{J} \end{aligned}$$

Actuator energy per exchange The actuator receives the PUT request and transmits the ACK:

$$\begin{aligned} E^{\text{CoAP}}_{\text{actuator},1} &= E_{rx}(70) + E_{tx}(15) \\ &= 70 \cdot 8 \cdot 58 + 15 \cdot 8 \cdot 50 \\ &= 32480 + 6000 = 38480\,\text{nJ} = 38.48\,\mu\text{J} \end{aligned}$$

Total energy per exchange

$$E^{\text{CoAP}}_{1} = 34.96 + 38.48 = 73.44\,\mu\text{J}$$

Total energy over 1 hour

$$E^{\text{CoAP}}_{\text{1h}} = 30 \cdot 73.44 = 2203.2\,\mu\text{J}$$

Therefore,

$$E^{\text{CoAP}}_{\text{1h}} = 2203.2\,\mu\text{J}$$

For completeness, the individual device consumptions are:

$$E^{\text{CoAP}}_{\text{sensor,1h}} = 30 \cdot 34.96 = 1048.8\,\mu\text{J}$$

$$E^{\text{CoAP}}_{\text{actuator,1h}} = 30 \cdot 38.48 = 1154.4\,\mu\text{J}$$

(b) MQTT via the gateway

Under the given assumptions:

- both devices start disconnected
- MQTT uses QoS 1
- connections are closed during sleep periods

So, for each measurement:

Sensor side The sensor performs:

- CONNECT: 60 B TX
- CONNACK: 50 B RX
- PUBLISH: 75 B TX
- PUBACK: 50 B RX
- DISCONNECT: 60 B TX

Hence:

$$\begin{aligned} E^{\text{MQTT}}_{\text{sensor},1} &= E_{tx}(60 + 75 + 60) + E_{rx}(50 + 50) \\ &= E_{tx}(195) + E_{rx}(100) \\ &= 195 \cdot 8 \cdot 50 + 100 \cdot 8 \cdot 58 \\ &= 78000 + 46400 = 124400\,\text{nJ} = 124.4\,\mu\text{J} \end{aligned}$$

Actuator side The actuator performs:

- CONNECT: 60 B TX
- CONNACK: 50 B RX
- SUBSCRIBE: 65 B TX
- SUBACK: 55 B RX
- receives PUBLISH: 75 B RX
- sends PUBACK: 50 B TX
- DISCONNECT: 60 B TX

Hence:

$$\begin{aligned} E^{\text{MQTT}}_{\text{actuator},1} &= E_{tx}(60 + 65 + 50 + 60) + E_{rx}(50 + 55 + 75) \\ &= E_{tx}(235) + E_{rx}(180) \\ &= 235 \cdot 8 \cdot 50 + 180 \cdot 8 \cdot 58 \\ &= 94000 + 83520 = 177520\,\text{nJ} = 177.52\,\mu\text{J} \end{aligned}$$

Total energy per measurement

$$E^{\text{MQTT}}_{1} = 124.4 + 177.52 = 301.92\,\mu\text{J}$$

Total energy over 1 hour

$$E^{\text{MQTT}}_{\text{1h}} = 30 \cdot 301.92 = 9057.6\,\mu\text{J}$$

Therefore,

$$E^{\text{MQTT}}_{\text{1h}} = 9057.6\,\mu\text{J}$$

For completeness, the individual device consumptions are:

$$E^{\text{MQTT}}_{\text{sensor,1h}} = 30 \cdot 124.4 = 3732\,\mu\text{J}$$

$$E^{\text{MQTT}}_{\text{actuator,1h}} = 30 \cdot 177.52 = 5325.6\,\mu\text{J}$$

Exercise Question 2 (EQ2)

I will choose **MQTT via gateway, with QoS 1, clean session=0, retained last temperature**. The broker stores the latest sample while the actuator sleeps, so only one delivery occurs every 30 min. Optimize freshness/successful delivery per joule; CoAP CON wastes energy in retries.